

SOUMYA SNIGDHA KUNDU

Mail · Website · Google Scholar · LinkedIn · X

EDUCATION

King's College London — Ph.D. Biomedical Engineering and Imaging Science	Oct 2023 – Oct 2027
Queen Mary University of London — M.Sc. Machine Learning (Distinction)	Sep 2022 – Sep 2023
SRM Institute of Science and Technology — B.Tech. Computer Science (Distinction)	Jul 2018 – May 2022

REPRESENTATIVE PAPERS

Hsieh, I., **S. S. Kundu**, Vercauteren, T., & Dorent, R. (2027). SinkSLOT: Sinkhorn via Sparse Lifted Optimal Transport. [AAAI](#).
S. S. Kundu et al. (2026). Instance Awareness of Multi-class Semantic Segmentation Loss Functions. [CV4CLINIC@CVPR](#).
S. S. Kundu et al. (2025). Cluster Dice. [SPIE Medical Imaging](#).
S. S. Kundu et al. (2024). Spinal osteophyte detection via robust patch extraction on minimally annotated x-rays. [IEEE ISBI](#).
Naidu, R. & **S. S. Kundu** (2021). Improved variants of Score-CAM via Smoothing and Integrating. [RCV@CVPR](#).
S. S. Kundu (2021). A Distributed Deep Learning Framework for Federated Big Medical Image Analysis. [IEEE Big Data](#).

RELEVANT INDUSTRY EXPERIENCE

Cosine (YC 23) — Machine Learning and Product Intern (Genie 3; Multilingual SWE-Bench)	Summer 2025
--	-------------

REPRESENTATIVE SOFTWARE & OPEN SOURCE CONTRIBUTIONS

Panoptica — [Maintainer] Instance-wise segmentation metrics	26★
Ultraflwr — [Author] Federated learning on the edge	20★
MONAI — [Core contributor] to the Medical Open Network for AI (MONAI)	8k★
PRs: Optax [#1340, #1366, #1458], batchgeneratorsv2 [#16, #19], torchsparsegradutils [#63, #65, #66, #67], OpenTUI [#253], Fresh [#421]	

REPRESENTATIVE FELLOWSHIPS & GRANTS

Google Summer of Code (GSoC) — Neuroinformatics Unit (1,141 of 23,371 proposals selected, ~4.9%)	2026
Claude for Open Source — Anthropic (open source maintainer grant, \$1,200)	2026
Codex for Open Source — OpenAI (open source maintainer grant, \$1,200)	2026
MRC DTP Postgraduate Studentship — King's College London (<2% selection, ~£205k)	2023
BDI Summer Internship — University of Oxford (1 of 4 selected)	2023
Summer Research Internship — IIT Gandhinagar (<0.008% selection)	2023
MITACS Globalink Research Internship — TRIUMF (UVic) (<3% selection)	2021

REPRESENTATIVE AWARDS

SPIE Medical Imaging Travel Grant Award (\$1000)	2025
IEEE ISBI'24 Best Student Poster Award Finalist (Top 8 of 717)	2024
UKRI Fast Start: Innovation Grant — Co-Applciant (~£50k)	2022

PROGRAMMING & SOFTWARE DEVELOPMENT

Languages: Python, C++, Bash, $\LaTeX$

ML: PyTorch, JAX, NumPy, CUDA

Tools: Git, Slurm, Docker, Kubernetes, macOS, Linux

GRADUATE TEACHING

Image and Signal Processing (Prof. James Housden)	Spring 2026
Python (Dr. Marc Modat and Dr. Jonathan Shapey)	Spring 2026
Journal Club — Biomedical Engineering (Prof. Monica Agromayor)	Winter 2024 & Spring 2025
Statistics — Biomedical Engineering (Prof. James Housden)	Spring 2025

SUPERVISION

Yingfang Tao — KCL BSc Thesis Supervision	2025–26
Jai Pradeep Awasthi and Pranav Rustagi (LNMIIT) with Yang Li	Summer 2025
In2Stem (4 High School students) with Lorena Garcia-Foncillas Macias	Summers 2024/25/26

ACADEMIC SERVICE

Reviewing: MICCAI, ICML, NeurIPS, ICLR, AISTATS, IEEE-ISBI.

References available on request.